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Molecular Constants of Lithium Hydrides by the Molecular-Beam Electric Resonance Method

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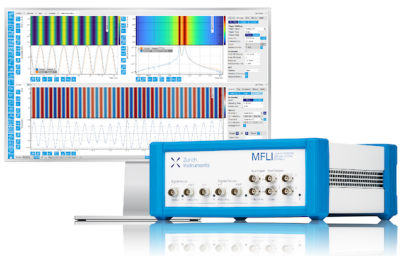
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Charge Carrier Mobility in Tetracene

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1 μsec or more depending on the discharge capacitor employed. The sample cell had a quartz window permitting excitation to 2000 Å, but for most measurement a 3300-Å interference filter was used.

The advantages of using an operational amplifier in the broad-band electrometer mode are: high input resistance (10^{11}), low noise level (1 mV), high gain, and high unity gain frequency. Current flowing through the sample is exactly balanced by current flowing through a feedback resistor (10^8 Ω), so that the amplifier input is held at virtual ground potential acting as a low resistance current meter. The principles involved have been described elsewhere.[4,6] Amplifier B was used in the integrating mode[5] where the integrating time was ~3 msec, and the charge sensitivity ~10^{-14} coulomb. The signal from either amplifier was fed to the oscilloscope which was triggered either internally or externally with an EG & G SGD-100 photodiode.

Tetracene was vapor zone refined in an apparatus similar to that described by Weisberg and Rosi,[7] and thin crystal flakes grown in a controlled sublimation furnace.[8] Electrodes were applied on the cleavage (ab) plane of the crystals, which ranged in thickness from 0.1–0.4 mm.

In order to assure the elimination of spurious effects, the mobility was determined by both the time-of-flight method[9] using amplifier A and the crystal counter technique[10] using amplifier B. Electrode materials were varied, both evaporated metal and blocking electrodes of thin mica or mylar layers being employed. In all cases, similar results were obtained. Hole mobilities measured at different temperatures are presented in Fig. 2. The electron mobilities were approximately the same, but the measurements were less reliable due to strong trapping. Hole and electron lifetimes were in the range 100–300 μsec.

The mobility of tetracene reported here of 10^{-2} cm^2 V^{-1}·sec^{-1} at room temperature is an order of magnitude lower than that calculated by Thaxton *et al.*[11] or estimated by Frankevich and Balabanov[12] but represents the first experimental values obtained for this material. Preliminary measurements have also been made on *p*-terphenyl crystals, and the hole mobility at room temperature is ~10^{-3} cm^2 V^{-1}·sec^{-1}, which

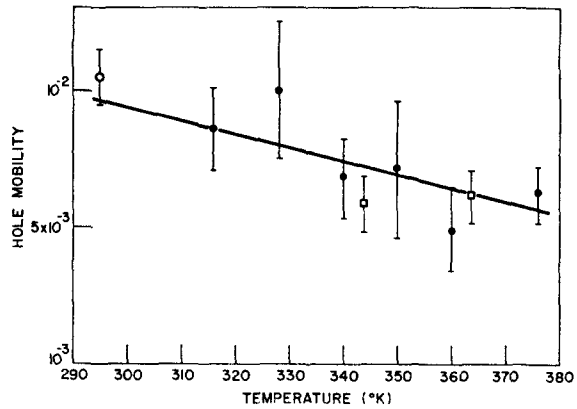


FIG. 2. Hole drift mobility (square centimeters volts^{-1}·seconds^{-1}) in tetracene crystals vs absolute temperature.

lies between the values reported by Mark and Helfrich[13] and Frankevich and Balabanov.[14]

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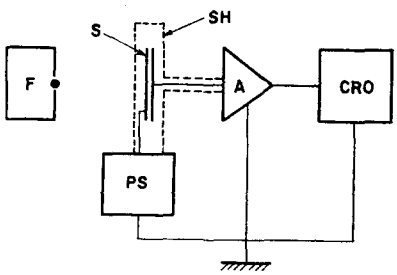


FIG. 1. Schematic diagram of apparatus. F, flash lamp; S, sample; SH, shield; PS, power supply; A, amplifier; CRO, oscilloscope.

Molecular Constants of Lithium Hydrides by the Molecular-Beam Electric Resonance Method*

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The molecular-beam electric-resonance spectroscopy of ^{7}LiH, ^{7}LiD, ^{6}LiH, and ^{6}LiD in the first rotational

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TABLE I. Molecular constants.[a]

Molecule	eqQ_D	eqQ_{Li}	c_D/c_H	c_{Li}	c_3	c_4	u
^{7}LiH $v=0$[b]		346.75(25)	−9.05(5)	10.025(75)	0.45318(100)	0.0(3)	5.8820(4)
^{7}LiH $v=0$[c]		346(1)	−8(1)	10(1)			5.882(3)
^{7}LiH $v=1$[b]		332.0(5)	−9.0(5)	9.8(1)	0.43828(100)	0.0(5)	5.9905(4)
^{7}LiH $v=1$[c]		328(4)	−6(2)	10(2)			5.990(3)
^{6}LiH $v=0$[b]		7.2(8)	−9.3(7)	3.9(5)	0.17111(100)	0.0(3)	5.8836(12)
^{6}LiH $v=0$[c]							5.884(3)
^{6}LiH $v=1$[b]		7.2(8)	−9.3(7)	3.9(5)	0.16588(100)	0.0(3)	5.9929(16)
^{6}LiD $v=0$[b]	33(1)	7.5(10)	−1.3(5)	3.2(10)	0.02639(100)	0.0(3)	5.8694(10)
^{6}LiD $v=1$[b]	33(1)	7.5(10)	−1.3(5)	3.2(10)	0.02577(100)	0.0(3)	5.9523(10)
^{7}LiD $v=0$[b]	33(1)	349(1)	0(1)	5.50(25)	0.05484(100)	0.0(3)	5.8677(5)
^{7}LiD $v=0$[c]	33(1)	349(1)		5.5(10)			5.868(3)

[a] Values of eqQ_D, eqQ_{Li}, c_D, c_H, c_{Li}, c_3, and c_4 are in kilocycles per second; value of u is in debyes.

[b] E. Rothstein, Ph.D. thesis, University of California, Lawrence Radiation Laboratory, Berkeley, Calif., Rept. UCRL-17948, February 1968.

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state and the lowest two vibrational states have been observed. Accurate dipole moments and nuclear–hyperfine interaction constants are determined (Table I). The Hamiltonian[1] used is

$$H=B\mathbf{J}^2-\tfrac{1}{2}\mathbf{U}\cdot\mathbf{E}-eq_1Q_1\left[\frac{3(\mathbf{I}_1\cdot\mathbf{J})^2+\frac{3}{2}(\mathbf{I}_1\cdot\mathbf{J})-(\mathbf{I}_1^2\mathbf{J}^2)}{2I_1(2I_1-1)(2J-1)(2J+3)}\right]$$
$$-eq_2Q_2\left[\frac{3(\mathbf{I}_2\cdot\mathbf{J})^2+\frac{3}{2}(\mathbf{I}_2\cdot\mathbf{J})-(\mathbf{I}_2^2\mathbf{J}^2)}{2I_2(2I_2-1)(2J-1)(2J+3)}\right]$$
$$+c_1(\mathbf{I}_1\cdot\mathbf{J})+c_2(\mathbf{I}_2\cdot\mathbf{J})$$
$$+c_3\left[\frac{3(\mathbf{I}_1\cdot\mathbf{J})(\mathbf{I}_2\cdot\mathbf{J})+3(\mathbf{I}_2\cdot\mathbf{J})(\mathbf{I}_1\cdot\mathbf{J})-2(\mathbf{I}_1\cdot\mathbf{I}_2)J(J+1)}{(2J-1)(2J+3)}\right]$$
$$+c_4(\mathbf{I}_1\cdot\mathbf{I}_2),$$

where u is the dipole moment, B the rotational constant, J the rotational angular momentum, E the Stark field, I_1 and I_2 the nuclear spins, c_1 and c_2 the spin–rotation constants, c_4 the electron-coupled spin–spin interaction, Q_1 and Q_2 the nuclear quadrupole moments, q_1 and q_2 the gradient of the electric field at nuclei 1 and 2 and $c_3=g_1g_2u_N^2(1/r^3)_{Av}$ with g_1 and g_2 as nuclear g factors and u_N the nuclear magneton.

When the values of $u_0(v=0)$ and $u_1(v=1)$ are fitted to the equation $u_v=u_e+u_I(v+\frac{1}{2})$; the u_e values for ^{7}LiH, ^{6}LiH, and ^{6}LiD are 5.82775, 5.82895, and 5.82795.

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Polarization of Balmer-Alpha Radiation from p + He Charge-Transfer Collisions*

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When a proton beam interacts with a target atom or molecule, the proton may capture an electron from the target to form a hydrogen atom in an excited state. If the excited hydrogen atom has a preferred orientation, the electromagnetic radiation subsequently emitted will be polarized. The magnitude of the polarization of the emitted radiation depends explicitly upon the electron-capture cross sections for capture into the various rotationally degenerate substates associated with a given configuration of the hydrogen atom. Formulas for the polarization of the emitted radiation are given by Percival and Seaton.[1]

Using a time-of-flight technique, Hughes, Doughty, and Filippelli[2] have reported the capture cross sections into the $n=3$ states of hydrogen for the charge-exchange collision of protons on N_2 molecules. In addition, they observed that the Balmer-alpha radiation from the $3D\rightarrow 2P$ transition was polarized to the extent of approximately 15% in the proton energy range $10<E<35$ keV. Also, Gaily, Jaecks, and Geballe[3] have reported that the Lyman-alpha radiation, produced in the charge-exchange collision of a proton beam with rare-gas atoms, was polarized. In the energy range $0.5<E<15$ keV, the polarization decreased monotomically from approximately 35% to 10%. In this note, we report upon the polarization of the Balmer-alpha radiation produced in the charge-exchange collision of protons on helium.

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